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OBJECTS

When AI Changes the Work We Sell

A Software Consultancy's Perspective

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andrena objects ag
AISoLA 2026 · Kos · 26 October 2026



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400 people, rooted in agile

WHERE WE COME FROM

Among the first that brought agile development into German organisations

> **“Agile is dead. Long live agentic.”**

WHO WE ARE TODAY



Our developers work inside client teams

Energy | Banking | Manufacturing | Retail | Mobility

Startups — Large corporations

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Software development is not coding

WHAT CLIENTS HIRE US FOR

PROBLEM

Understand the problem and its risks

QUALITY

Hold a quality bar under feature pressure

CHANGE

Keep the system maintainable

TEAM

Improve how the client team works

Engineering makes a system useful today and changeable tomorrow.

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We only make sense where value concentrates

FOR EXAMPLE	WHAT THE WORK INVOLVES
A power grid operator	Large, complex systems
A railway operator	Unclear requirements
A central bank	Deep domain knowledge
IT providers that supply software to hundreds of banks	An IT landscape we have to fit into

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AI changes our tasks, not our purpose

Purpose
unchanged

Create value for the client by managing complexity under change

so far

Tasks
changing radically

How we specify, build, check and decide

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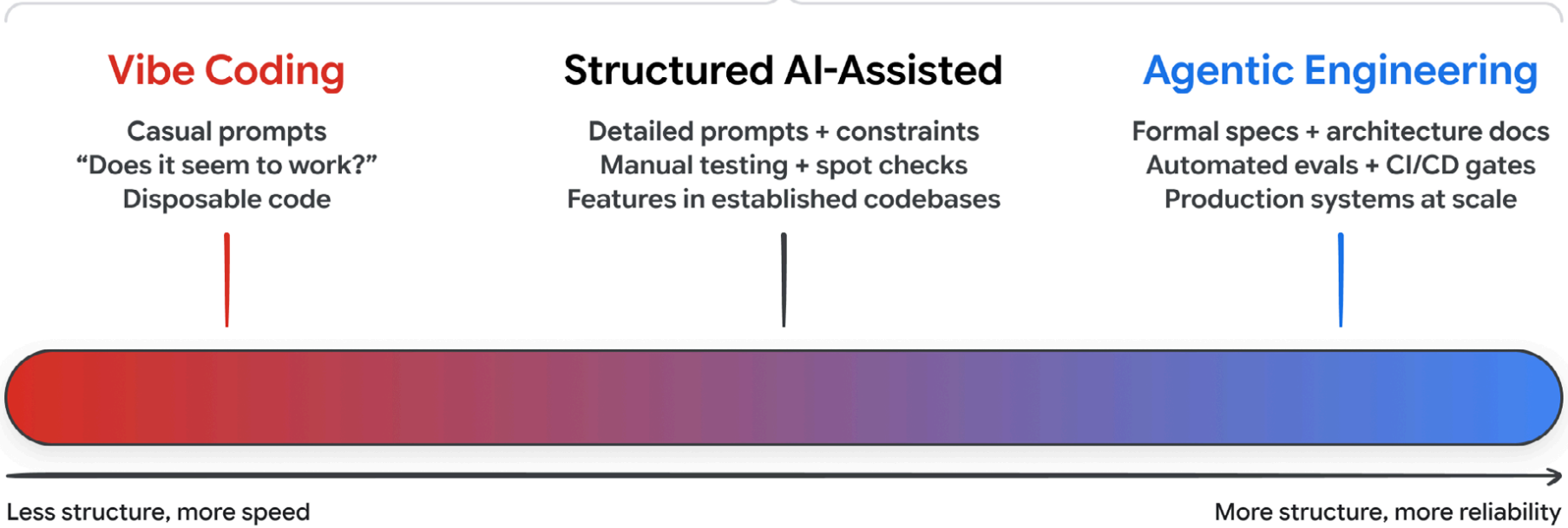
Impact of AI on our work

Dynamics and mental models

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From vibe coding to agentic engineering

The differentiator is not whether you use AI — it's how outputs get verified.



The right position depends on the stakes.
The skill is knowing where to draw the line for each task.

Source: Osmani, Saboo & Kartakis · The New SDLC With Vibe Coding · Google, May 2026 · Figure 3, p. 14

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Most of our work is agentic engineering

Vibe coding

less structure, more speed

Agentic engineering

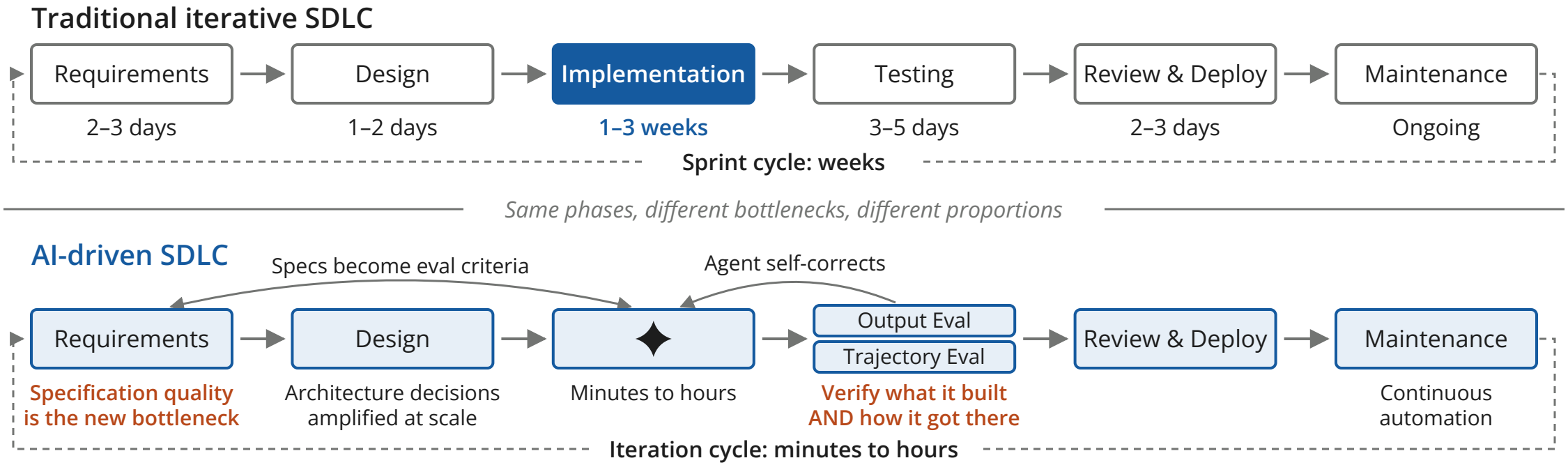
more structure, more reliability



Spectrum after Osmani, Saboo & Kartakis, 'The New SDLC With Vibe Coding' (Google, May 2026)

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The bottleneck moves to specifying and checking



Developers move from producing code to supervising it. Responsibility stays with people.

After Osmani, Saboo & Kartakis, 'The New SDLC With Vibe Coding' (Google, May 2026), Figure 5; emphasis added · pattern: Goldratt, The Goal (1984)

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Autonomy levels: who approves what

L5	**Dark factory**	person sets guardrails
L4	**Software factory**	person is auditor

Up to L3, teams often get there on their own; from L4, it takes decisions beyond the team

L3	**Delegated agent**	person is change owner
L2	**Conversational**	person approves each change
L1	**Assistive**	person decides everything
L0	**No AI**	person does all the work

Working tool from our transformation concept (Oct 2026), not validated · Ladder adapted from ASDLC ‘Levels of Autonomy’ and Swarmia

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L3 within the team, L4 beyond it

L5 Dark factory	Dark factory: one continuous loop						
L4 Software factory	Inception			Construction			Operation
L3 Delegated agent	Delegate the analysis	Spec → backlog	Delegate the design	Build → test → ship loop			Incident → fix PR
L2 Conversational	Co-draft the case	Co-write the PRD	Co-design	Code & test together		Co-build pipelines	Debug together
L1 Assistive	Ask & brainstorm	Phrase a story	Design advice	Autocomplete	Write a test	Draft the config	Explain the error
L0 No AI	Manual research	Hand-written specs	Whiteboard design	Hand-coded	Manual testing	Manual releases	Manual ops
	STAGE 1 Strategy & Discovery	STAGE 2 Requirements Engineering	STAGE 3 Architecture & Design	STAGE 4 Implementation	STAGE 5 Quality & Testing	STAGE 6 Deployment & Release	STAGE 7 Operations & Maintenance

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One team: the level differs per stage

L5 Dark factory	Dark factory: one continuous loop						
L4 Software factory	Inception			Construction			Operation
L3 Delegated agent	Delegate the analysis	Spec → backlog	Delegate the design	Build → test → ship loop			Incident → fix PR
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	STAGE 1 Strategy & Discovery	STAGE 2 Requirements Engineering	STAGE 3 Architecture & Design	STAGE 4 Implementation	STAGE 5 Quality & Testing	STAGE 6 Deployment & Release	STAGE 7 Operations & Maintenance

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The downside of delegation

Handing off
intended, cognitive offloading

You hand routine work to the agent and work on harder problems.

→

Deskilling
skills decline

Skills you stop practising decline, including the skill to check.

→

Cognitive debt
understanding decays

The system works. Nobody in the team can explain why.

→

Cognitive surrender
judgement given up

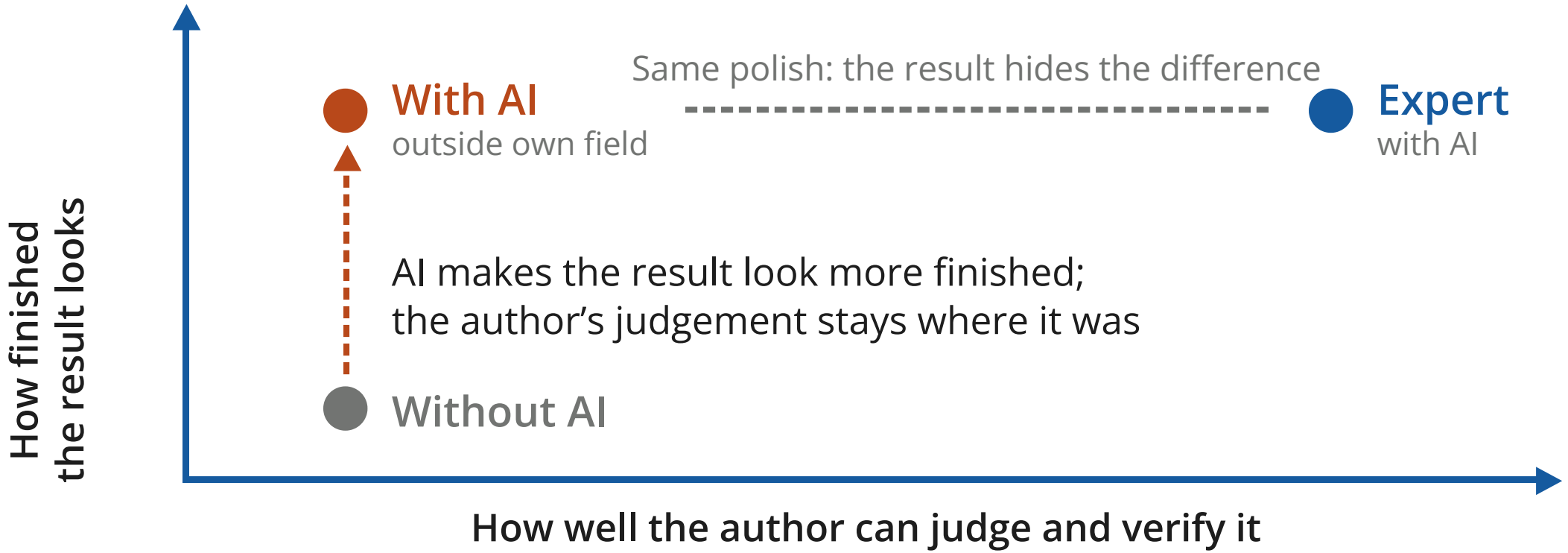
Plausible output is accepted without checking.

> **Handing off routine work is intended. It becomes a problem when nobody in the team can still check the result.**

Cognitive surrender: Shaw & Nave, Wharton working paper (2026) · Bainbridge, 'Ironies of Automation' (1983)

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Polished output hides missing judgement



'Output–competence decoupling' after 'Appearing Productive in the Workplace' (nooneshappy.com), 'The conduit problem'

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AI-written code keeps getting better

FrontierCode

Coding benchmark by Cognition, 2026

Easy tier: 50 everyday coding tasks, solved in one shot

A patch passes only if it meets the maintainers' merge criteria, not just the tests

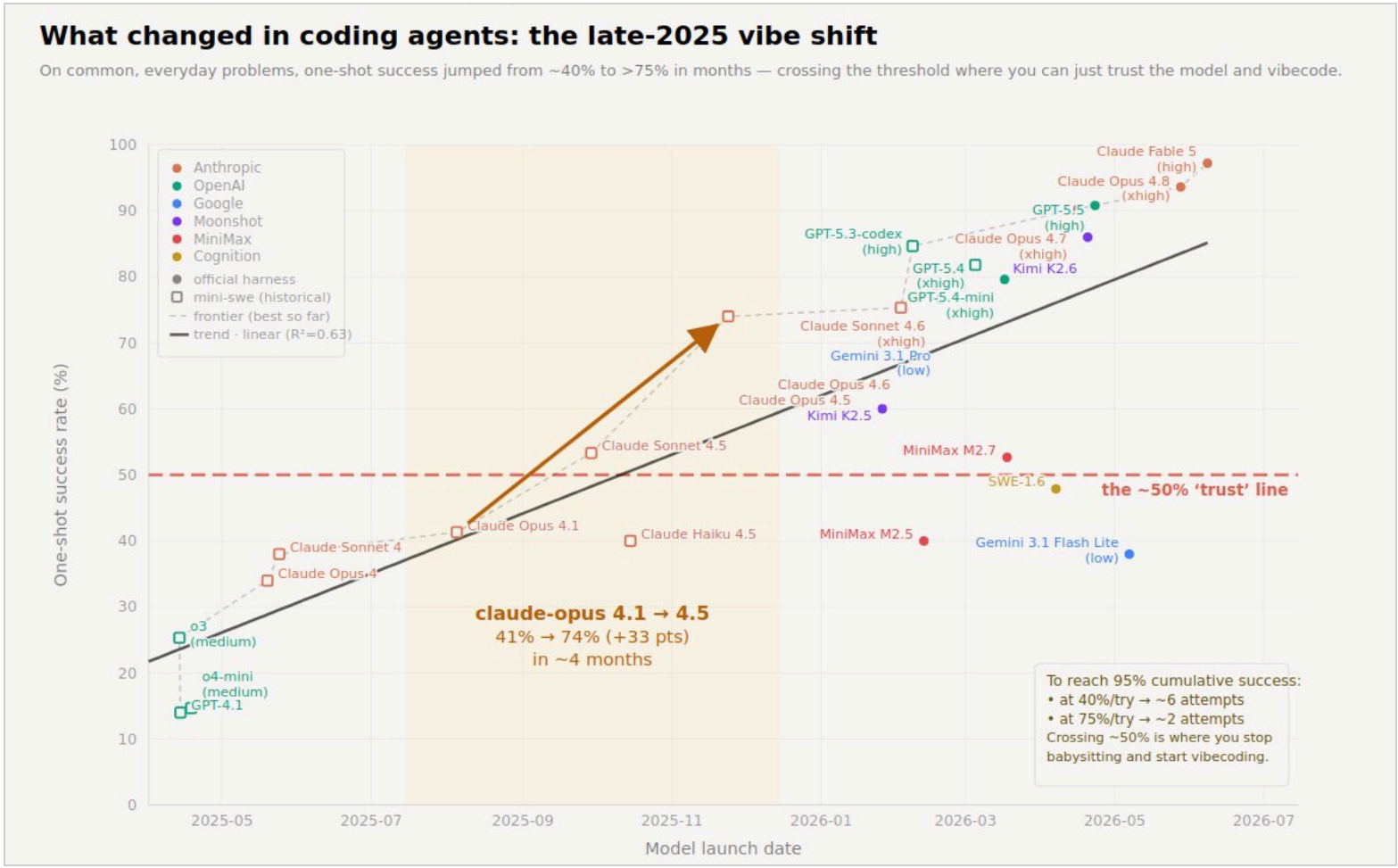
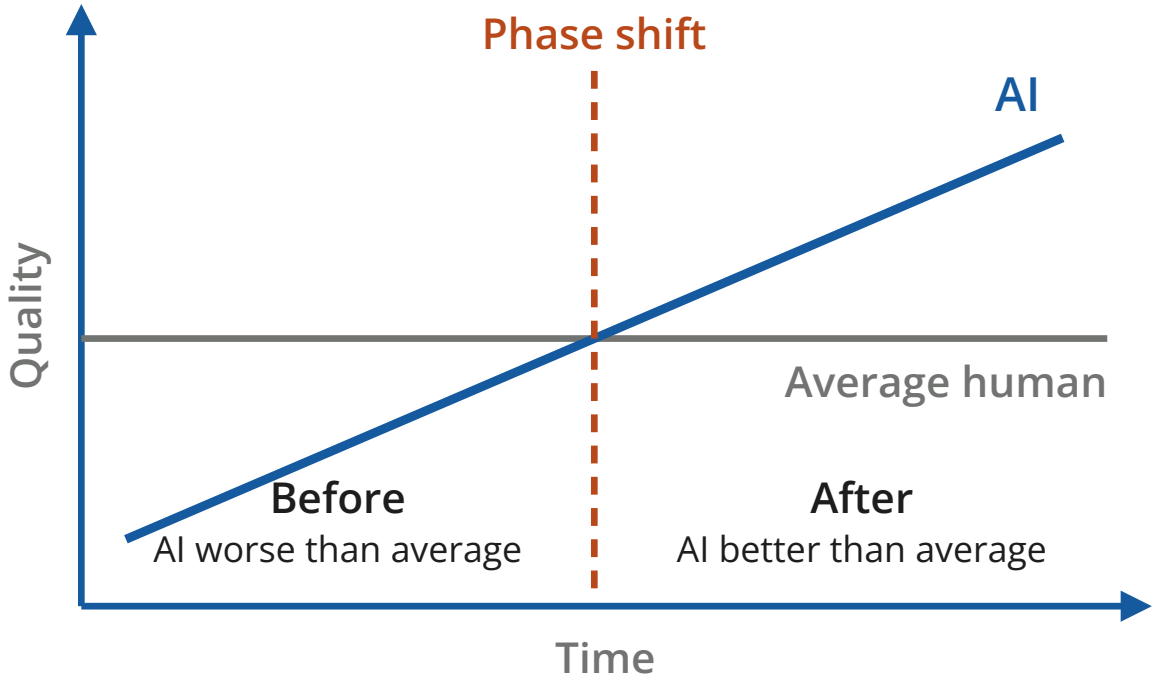


Chart by @swyx from FrontierCode (Cognition) results, Easy tier

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Did the phase shift already happen?

The trend suggests a phase shift



Some believe it has happened

> The second criterion is writing code that another engineer can understand and build on. Here the gap between humans and AI persists, but is closing fast. There isn't full consensus among staff at Anthropic, but many believe that the Claude-written code was still worse in quality than human-written code at Anthropic in late 2025, and is roughly at parity today. We expect it to be better within the year.

Anthropic Institute, 'When AI builds itself'

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A more nuanced view of the phase shift

Terms coined by John Ousterhout in *A Philosophy of Software Design* (2018)

TACTICAL PROGRAMMING

Make this change work now

● Phase shift: done

STRATEGIC PROGRAMMING

Keep the system changeable

○ Phase shift: not yet

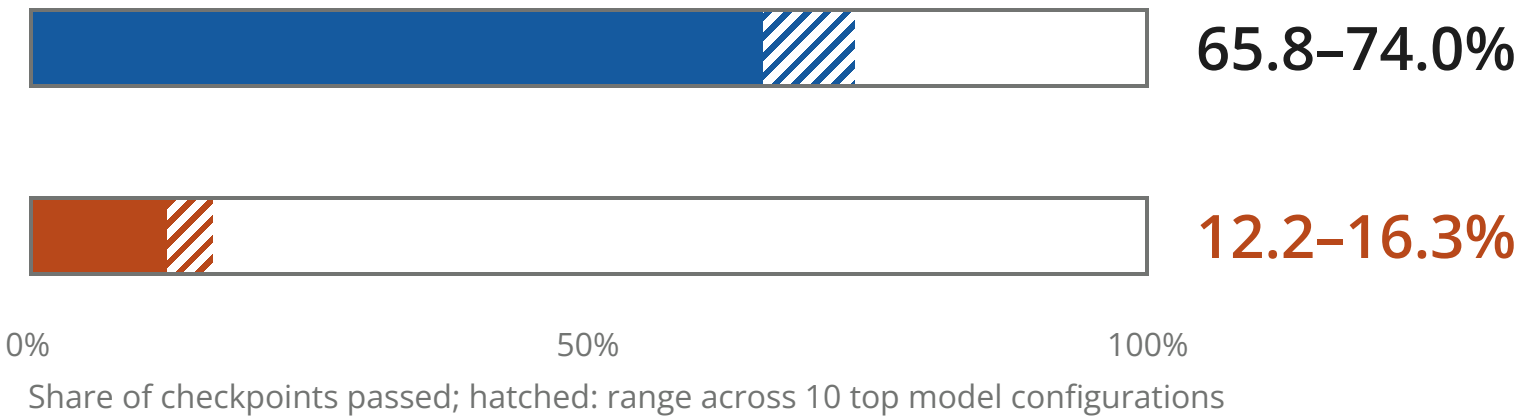
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Strategic programming: no phase shift yet

SlopCodeBench: an agent adds a series of new requirements to its own code

Core pass
core tests of the checkpoint pass

Strict pass
all tests pass, including earlier features

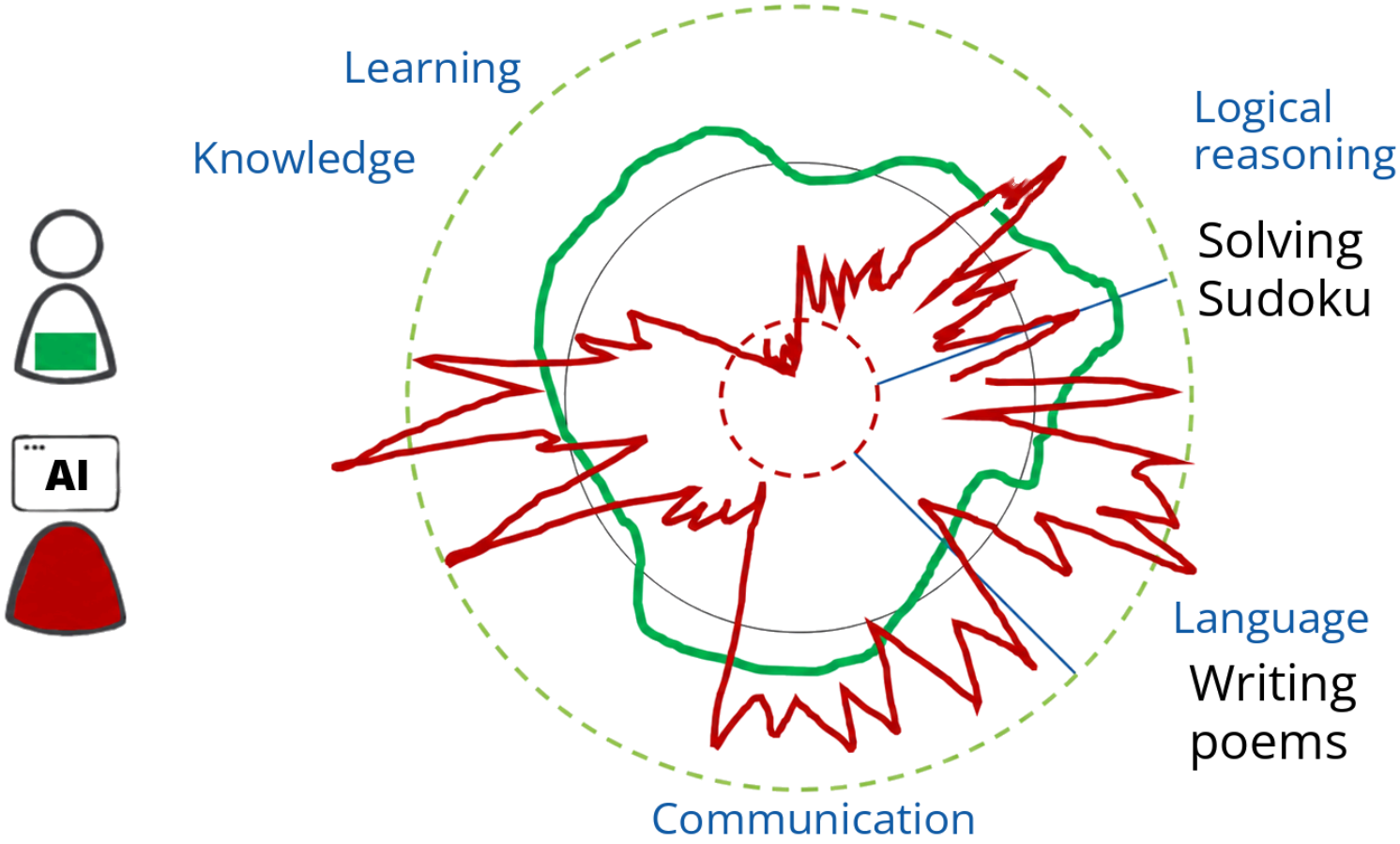


> **Agents often build the next feature but much less reliably keep earlier features working.**

SlopCodeBench; independent community evaluation by Dex Horthy (HumanLayer), 15 Oct 2026, one run per configuration

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LLMs ≠ humans: the jagged frontier



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The profile changes unpredictably

Mathematics as a warning

Jul 2025 — **Gold-medal score at the Mathematical Olympiad**
5 years before the median expert forecast

Sep 2026 — **Claimed solution of a Millennium Prize problem (Navier–Stokes)**
By OpenAI, not confirmed

Oct 2026 — **Proof of the quasi-Riemann hypothesis**
Experts largely agree, review continues

Low-hanging fruit in the model labs

Post-training

Test-time training

Agent coordination

AI-assisted research

Not: more data, larger models, more compute

Sources: Forecasting Research Institute (Sep 2026); Wikipedia, Navier–Stokes (Oct 2026); Quanta Magazine (Oct 2026)

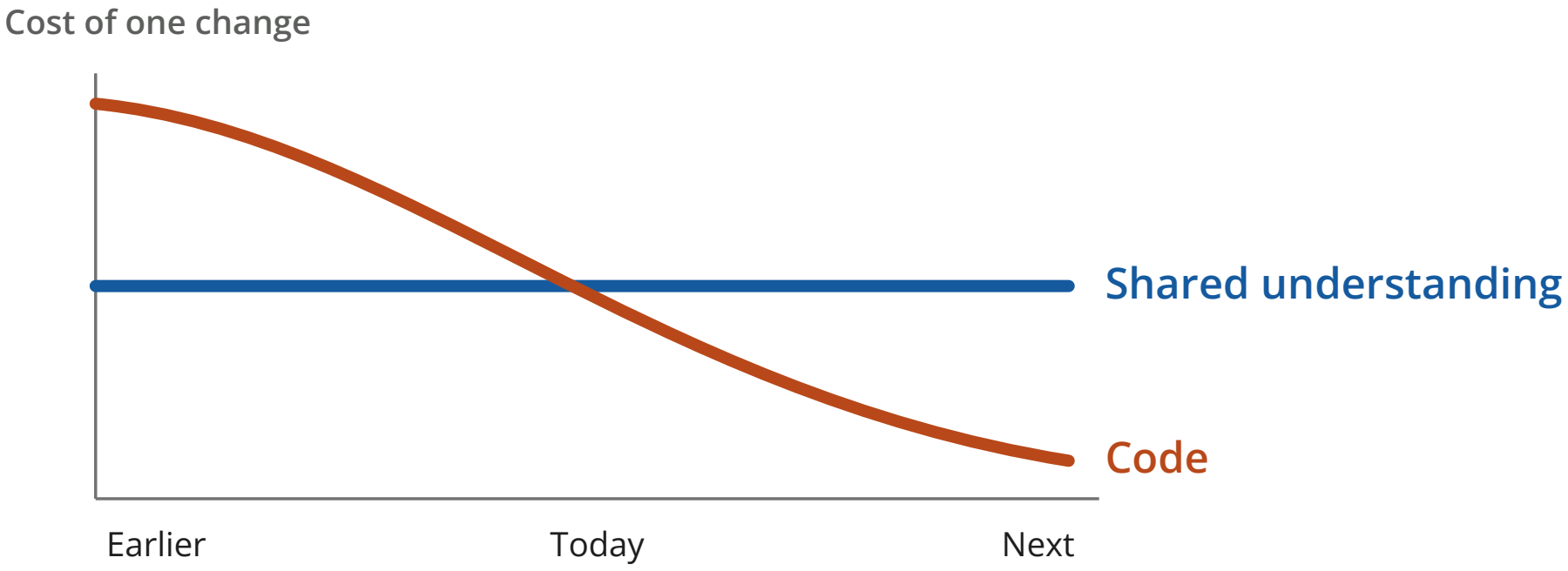
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Today's shortcomings

- Saying no
- Architecture
- Making risks transparent
- Owning the outcome

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Engineers are still needed



AI lowers the cost of code faster than the cost of a shared understanding of the system.

Related: Peter Naur, ‘Programming as Theory Building’ (1985)

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Five patterns at our clients

1. Use of AI spread early; delegation to agents spreads unevenly.
2. Costs are visible; value is not.
3. Checking limits delegation.
4. The gap between what agents can do and what teams use is growing.
5. The usual blockers: mindset, mental models and the organisation itself.

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Most of our developers use AI daily

USE AI DAILY OR ALMOST DAILY

66%

Nov 2025



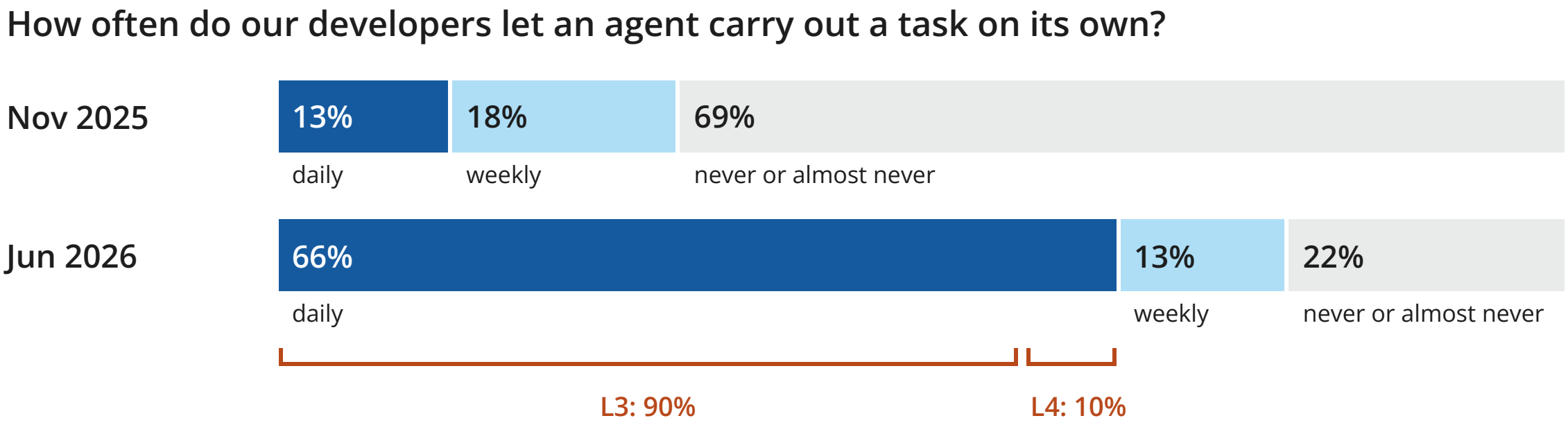
86%

Jun 2026



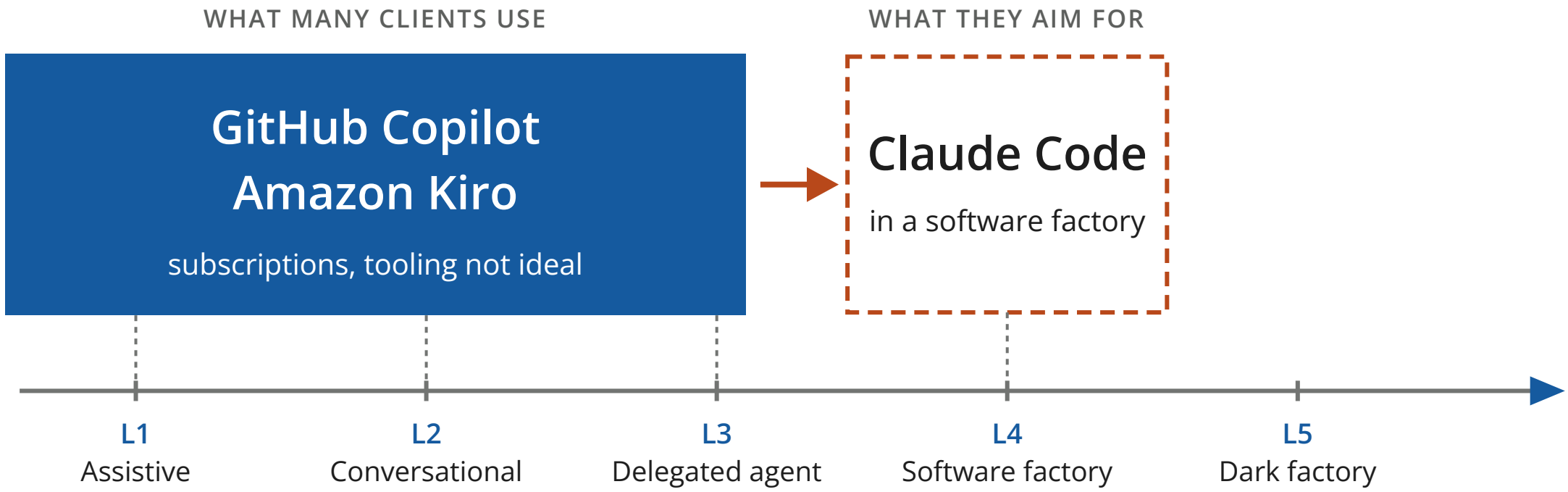
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Delegation to agents varies widely



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Clients aim for L4, work at L1–L2



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Costs are visible, value is not

AS A RESULT

Absurdly small budgets

$20–60

per user and month

→

Pays for itself with

1 hour

saved per month: very easy

→

Often even easy

10×

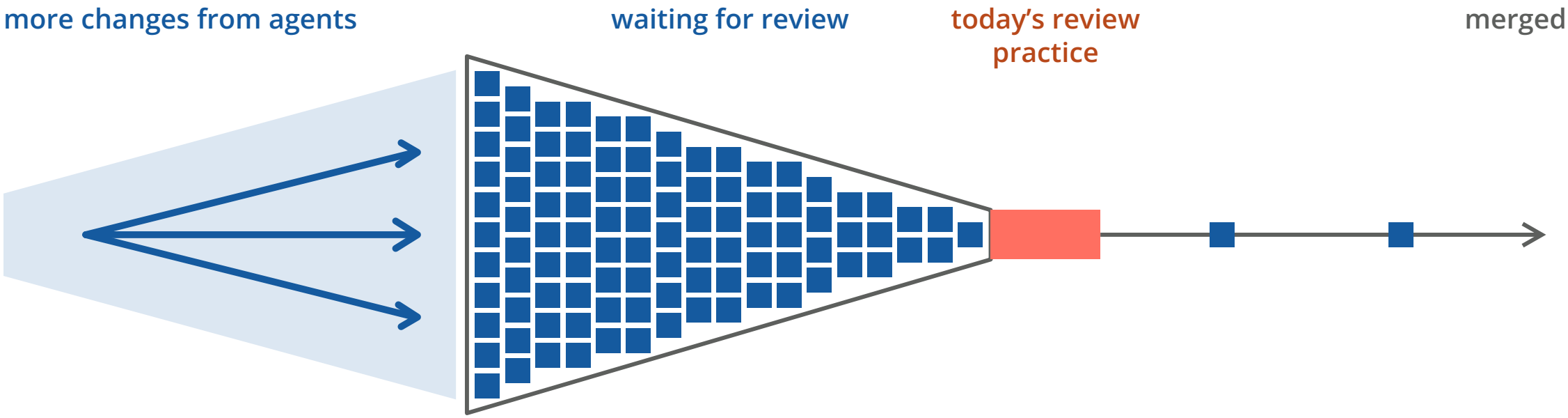
saving per AI dollar

Output is not outcome



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Code review is by far the biggest barrier



Keeping today's review practice while throughput rises leads to a lot of frustration, in the best case.

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Rebuild review from its purposes

STEP 1

Discuss what review is for in our team

- Find bugs
- Check intent and design
- Keep code maintainable
- Spread system knowledge
- Develop judgment
- Assign responsibility
- ...

STEP 2

Achieve each purpose with AI in mind

Move to tests, tools and AI review

for example conventions, regressions

Still need deliberate human work

for example system knowledge, responsibility

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Achieve each purpose with AI in mind

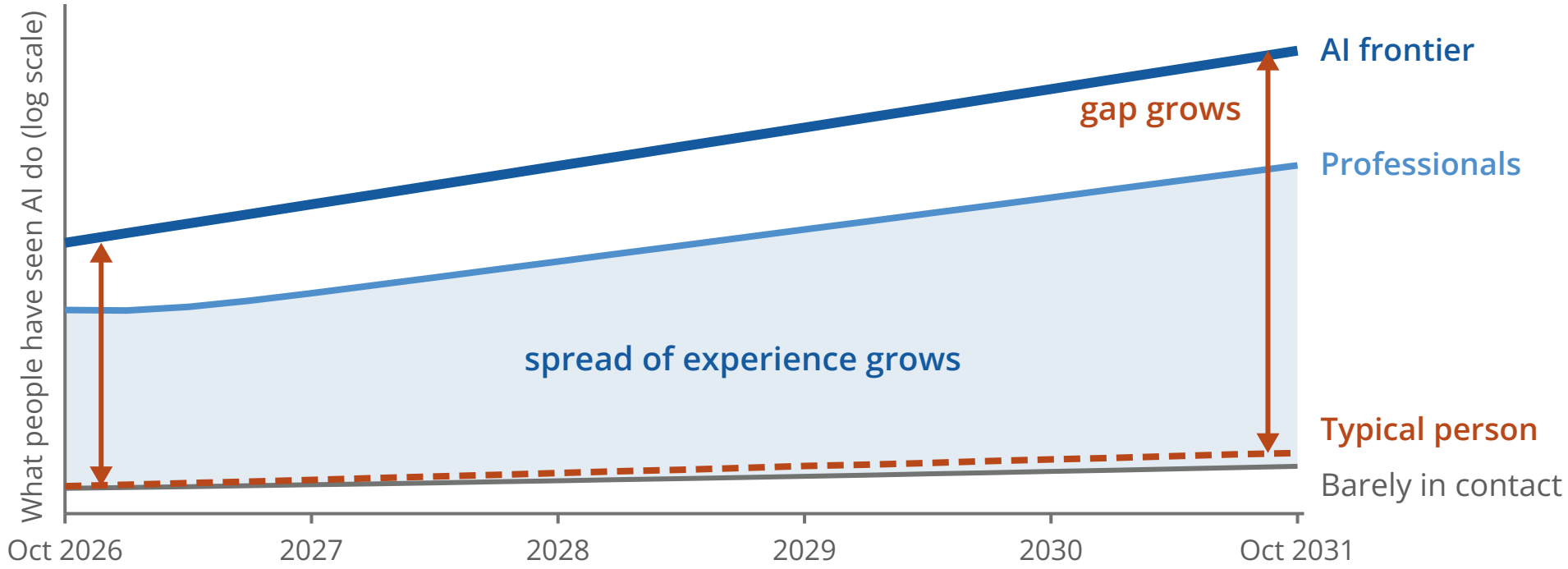
PURPOSE	PRACTICE WE RECOMMEND
Find bugs	Read Anthropic's blog post on code review: claude.dev/blog/solving-code-review-at-anthropic
Check intent and design	Review on a higher level of abstraction, not line by line
Spread system knowledge	Regular mob sessions: the team explains and reasons about one subsystem together
Assign responsibility	Name who accepts the risk and who handles failures
…	…

Osmani, Lunt, Wang, claude.dev/blog, 15 Oct 2026

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The understanding gap is growing

Advancement: the frontier doubles about every 7 months **Upskilling:** people catch up far more slowly



Steve's simulation of Karpathy's funnel (Oct 2026), base scenario; illustrative model, not a forecast

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Liberation or loss?

BLOCKER 1 · MINDSET

Crafters

love writing code

They grieve the craft of writing code.

Builders

love the result

They still get the satisfaction when it works.

After Les Orchard, ‘Grief and the AI Split’, blog.lmorchard.com (11 Mar 2026)

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Which joy is yours?

BLOCKER 1 · MINDSET

Addy Osmani: most of us draw on three joys, mixed differently. Agents affect each one differently.

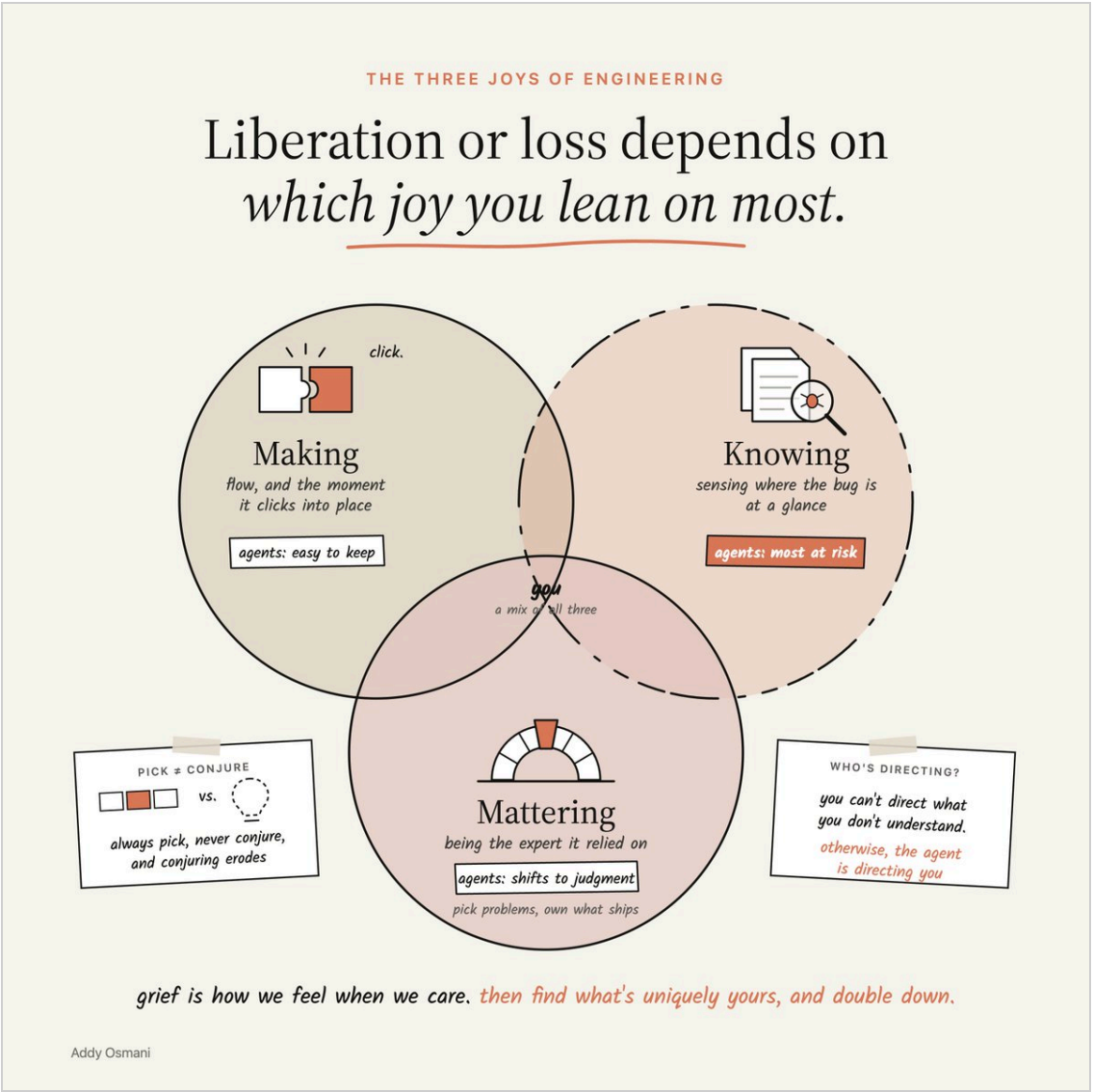


Illustration: Addy Osmani, post on X (9 Oct 2026)

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Mental models are hard to update

BLOCKER 2 · MENTAL MODELS

Examples of unexamined beliefs

‘Writing code is the core job function.’

‘If it was fast, it cannot be valuable.’

‘Every line must be manually reviewed.’

without asking which purpose that serves

Building mental models is my job as a consultant and yours in research.

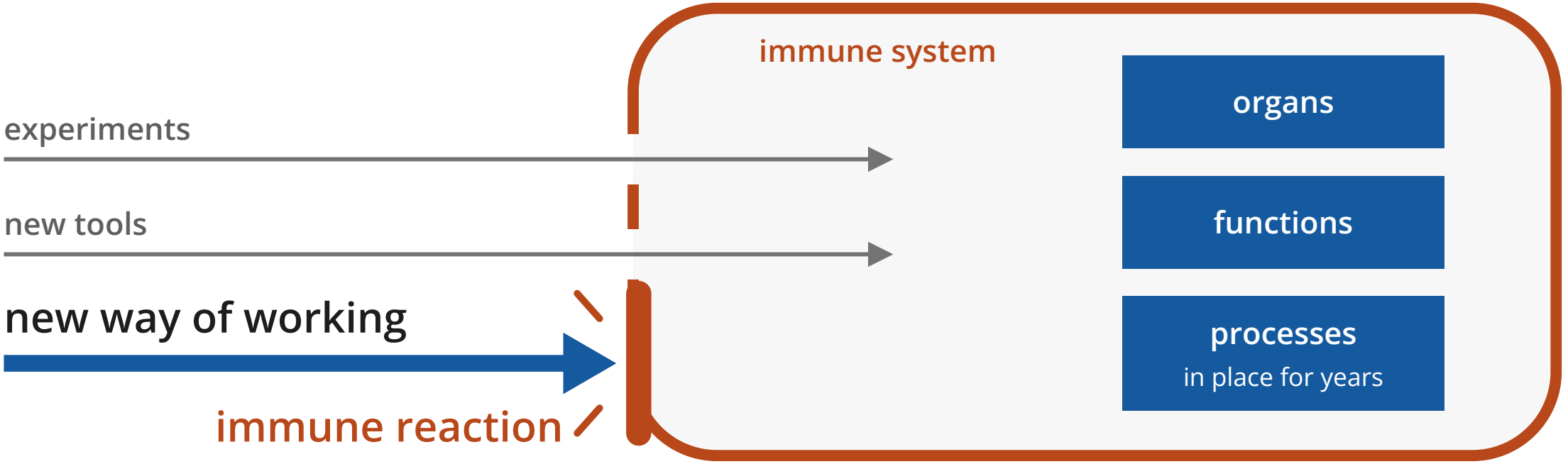
> **Newcomers can overtake experienced people like us.**

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Organisations have an immune system

BLOCKER 3 · ORGANISATION

Organisations readily try experiments and new tools. Changing a way of working they have used for many years triggers an **immune reaction**.



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How we respond

1. Strategic thought
2. Training: upskilling our existing engineers
3. The junior problem
4. Questioning best practices
5. New business models

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Plan for the slope

OUR PLANNING RULE

Plan for the slope, not for today’s profile.

AI CAPABILITIES

Phase shifts are sudden.

OUR SIDE

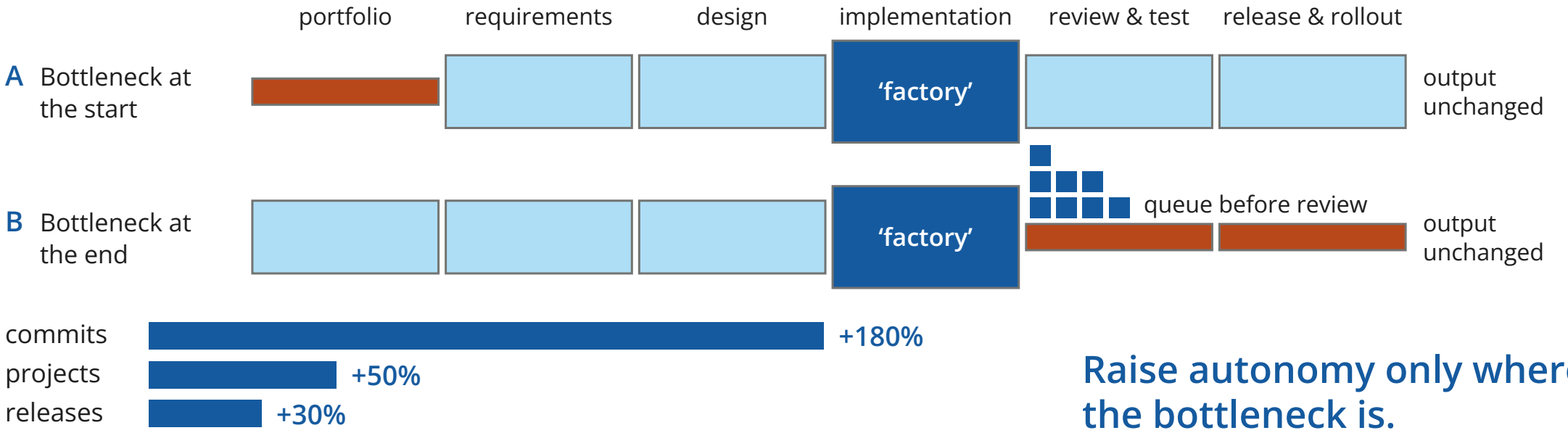
Changing roles, skills and organisations is slow.

So we start early.

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Raise autonomy at the bottleneck

TWO PATTERNS (ILLUSTRATION)



After adopting autonomous coding agents; >100,000 GitHub developers; NBER WP 35275 (2026)

Raise autonomy only where the bottleneck is.

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Training our existing engineers

2023 – end of 2025

AI basics course

One day, voluntary

Early 2026

Redone one-day training

Migrating code with Claude Code; main takeaway: compounding engineering

Mandatory: 90% of engineers in under six months

Since October 2026

Modular approach

Answer to the widening gap

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AI training is not enough

ENGINEERING EXPERTISE

We develop colleagues into engineering experts

We train and upskill them beyond AI. Only expertise can judge output that looks finished.

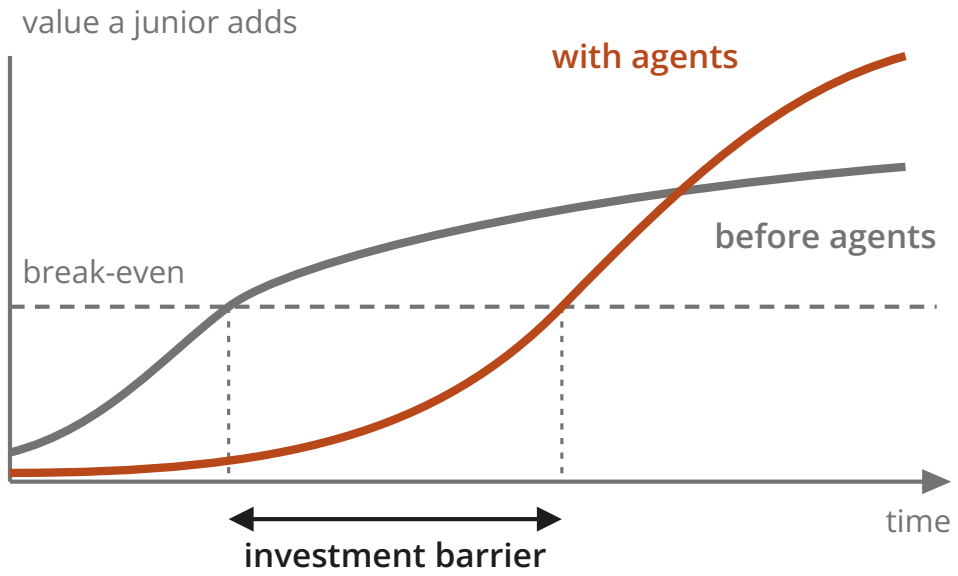
PRODUCT SIDE

The role shifts towards a product developer

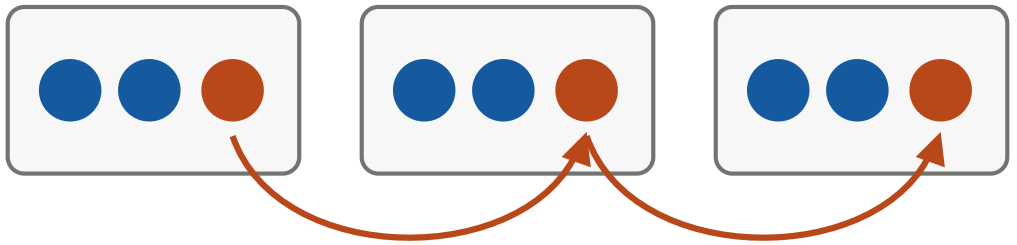
More responsibility across the SDLC. Our seniors already work this way.

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The junior dilemma



Sketch, no data: a junior takes longer to cover their cost



experienced junior, cycles through the pods

Our idea, not yet evaluated: pods of three, always two experienced people and one junior; the junior moves on to the pod with the most promising idea

> Software engineering moves from overnight experts into pilot or physician territory: years of training first.

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Questioning established practices

? **Code review**

? **Unit tests**

? **Test pyramid**

? **XP practices**

Pair programming | TDD | Refactoring | ...

? **Clean code**

Meaningful naming | Small functions | SRP | DRY | KISS | YAGNI | Separation of concerns | Low coupling | High cohesion | ...

> **Our stance:** principles stay, thresholds change.

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Day rates punish productivity

TIME AND MATERIALS



The client keeps the whole gain. Slow work earns us more.

WHAT WE FAVOUR

Fixed price

Where outcome and acceptance criteria are clear and checkable

Per story point

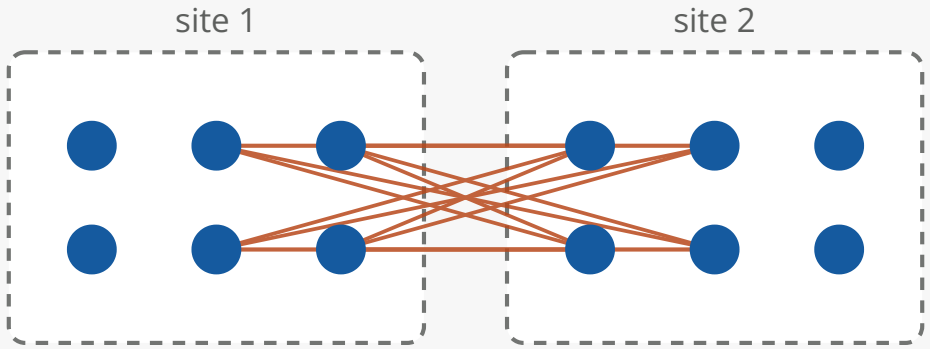
Price follows delivered scope

New: offered by only a few providers

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Fighting nearshoring with tiny AI teams

NEARSHORING OR OFFSHORING

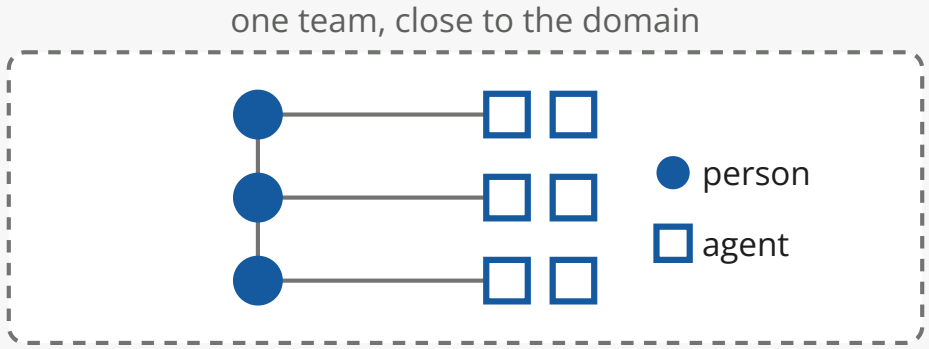


Larger team, big communication overhead

About 30% of the effort goes into coordination (our estimate)

VS

TINY AI TEAM



Three people with AI

Highly productive, without the communication overhead

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Where research can help us the most



1

Dependable delegation

2

Measuring what agents contribute

3

Developing judgement

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1 Dependable delegation

Clients want L4. Human review is already the bottleneck.

What evidence justifies releasing changes without human approval?

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2 Measuring what agents contribute

Clients can count AI costs. They struggle to quantify the gains.

How do we measure AI's contribution to client value?

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3 Developing judgement

AI takes over the tasks juniors used to learn on.

How do we develop judgement when AI does the implementation?

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Contact & questions

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SLIDES AS PDF

github.com/haupts/aisola-slides-2026

> **Thank you. Questions welcome.**

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